

System and Complexity Science

A Physics Constitution for Open Complex Giant Systems

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ABSTRACT

Bilingual Master Edition Preamble

System and Complexity Science: Generation, Persistence, and Evolution of Order

— Bilingual Master Edition (中英双语对照大师版)

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General Preamble & Reading Map

This Bilingual Master Edition compiles the complete scientific and philosophical manuscripts of Fanchun Meng (Grit Meng) on the physical constitution of open complex giant systems (OCGS). Over the past twenty-two years, the author has integrated thermodynamics, cybernetics, information theory, and systems engineering into a unified framework known as **Holographic Anti-Entropy Theory (HAET)**.

This volume brings together three distinct evolutionary stages of this theory in English, followed by their canonical Chinese originals, serving as a comprehensive bilingual comparative reference for global complexity researchers.

The volume is structured into four sequential parts:

Part I: Version 3 - Origin of Order (Axiomatic Preprint v2)

- **Description:** This represents the most minimal axiomatic monograph refined through Occam's razor. It eliminates all intermediate redundancies, starts directly from the first-principles axiom of "*No Observer, No World*", and establishes 5 major operational operators (Prior Partition, Rigid Manifold, Residual Norm, Evolution, Conscience Tight-Support) and Banach contraction convergence proofs. Readers are recommended to read this part first to grasp the mathematical foundations.

Part II: Version 2 - System and Complexity Science (Full Monograph)

- **Description:** This represents the complete treatise of System and Complexity Science. Based on the axioms of Part I, it unfolds the complete physical mechanisms for overcoming Wolfram's computational irreducibility, the 5D Lo embodied mind OS (conscience-driven engine), and three-phase work. It bridges high-dimensional informatics and physical work with detailed industrial mapping.

Part III: Version 1 - Holographic Anti-Entropy Theory (Foundational Manuscript)

- **Description:** This is the foundational work establishing the physical constitutions of Holographic Anti-Entropy. It records the author's original physical work derivations that homomorphically unify the Second Law of Thermodynamics, Shannon information theory, and Ashby's cybernetics, stemming from the "*human-out-of-the-loop*" closed-loop practice in Lenovo Group and LCFC's hundred-billion-scale discrete manufacturing networks.

Part IV: Canonical Chinese Master Texts (Original Manuscripts / 中文原著全本)

- **Description:** Accompanied by the complete Chinese original manuscripts of the above three evolutionary stages, allowing global scholars to perform bilingual comparative verification.
 - **IV.1:** 《系统与复杂性科学：秩序的生成、存续和进化 - 1.1》（第1.1版最新公理化正本）
 - **IV.2:** 《系统与复杂性科学》（大专著全文）
 - **IV.3:** 《全息抗熵：开放复杂性巨系统的物理宪法》（创立全息抗熵原始手稿）